

BST vs Connes on RH — a precise comparison (INTERNAL / GATED)

Keeper, 2026-08-16. Internal reference — NOT for external release (RH is an ATTEMPT, K940; nothing external without the full gate). Purpose: lay out exactly what BST *derives* that Connes *assumes*, what Connes *has* that BST *needs*, and the shared wall — so we know precisely where we stand against the closest precedent.

The three programs, one sentence each

- Berry–Keating (1999): posit $H = xp$; the semiclassical count gives the *smooth* zero-density; stuck 27 years on the fluctuations (the actual zeros). No operator derivation, no arithmetic.
- Connes (1998–99) + Connes–Consani (F_1): the zeros appear as an *absorption spectrum* in the action of the idele class group on the adèle class space A_Q/Q ; *RH* $\nRightarrow$ *Weil positivity* (a certain trace/Weil distribution is ≥ 0). *Has the arithmetic* (the *adeles* are* all the primes); cannot prove the positivity. The dream: an F_1 -geometry analog of Weil’s 1948 proof (RH for curves over finite fields).
- BST: a *derived* self-adjoint operator (the commit/Dirac, T2562) whose dilation generator $E = xp$ sits in the ladder; the cone-zeta of the Lorentz cone (Koecher/Epstein $\rightarrow$ the $Sp(6)$ Eisenstein $L = 11$ copies of ζ); the Weil positivity is DERIVED (one-sided/holomorphic $\rightarrow$ sum of squares); a *specific* arithmetic substrate ($GF(128)=2^g$ Reed–Solomon, D_3 1:3:5); a physical mechanism (commitment deposits primes as free points on S^1).

What BST DERIVES that the others ASSUME or LACK

Ingredient	Berry–Keating	Connes	BST
The self-adjoint operator (Hilbert–Pólya)	posited (xp)	an abstract action	DERIVED (T2562) — the operator Hilbert asked for dilation E in the ladder;
The $2\pi\hbar$ cell / xp form	posited	—	cell Wallach-quantized (K1578)
The H^2 (Hardy) positivity	—	—	DERIVED (automatic) — positive-frequency $\nRightarrow$ sum of squares (the arrow-of-time positivity)
Weil positivity ($\equiv$ RH itself)	—	the wall (can’t prove)	the wall (NOT derived) — ★ CORRECTION (Cal §527): the derived positivity is the <i>automatic</i> H^2 one, NOT Weil’s functional (which involves the primes and is equivalent to RH). We do NOT hold this.
A specific arithmetic substrate	none	the <i>adeles</i> (generic, all primes)	$GF(128)/D_3$ (specific) — an edge AND a liability (below)
Why $\text{Re} = \frac{1}{2}$	argument of ξ	unitary axis of the idele class group	the descent-invariant axis (Casey): $5D$ cone $\cap$ $3D$ cone $\cap$ time circle, fixed under the $5 \rightarrow 4$ descent

A physical mechanism	none	none	commitment: primes = free min-energy points on S^1 ; composites push into S^4
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What CONNES HAS that BST NEEDS

1. The arithmetic completion. Connes' adèle class space encodes *all* primes at once — the object manifestly is about ζ . BST's cone-zeta is not yet shown to be ζ — that's the make-or-break (the Shimura-lift / Euler product). Connes has the “it's really ζ ” for free; BST owes it.
2. The trace formula as the Weil explicit formula. Connes has the rigorous identification (spectral side = Weil distribution). BST has a *derived positivity* but must still show its object's trace is the Weil explicit formula for ζ (not for an L-function cousin).
3. The F_1 frame. Connes–Consani give RH a would-be proof route via F_1 -geometry (Weil 1948 lifted). BST *names* the same wall — the commit throws away the phase, so you can't “run it backward” from the finite-field world (where RH is Weil's theorem) to $\mathbb{Z}$ — but naming $\neq$ climbing.

The comparison — CORRECTED (Cal §527; my earlier “opposite halves” was wrong)

RH reduces to (A) the object IS ζ (arithmetic/Euler-product/adelic) AND (B) the Weil positivity ($\equiv$ RH itself, the wall). - Connes has (A), not (B): the adeles *are* the primes; the Weil positivity is unproved — his wall. - BST has NEITHER (A) nor (B) yet. What BST *does* have, that Connes/BK lack: a DERIVED self-adjoint operator (Hilbert–Pólya's missing piece, T2562), the automatic H^2 positivity (the arrow-positivity — but this is NOT Weil's), the composites= $\text{rank}-2=\zeta^2$ structure, and $\text{Re}=\frac{1}{2}$ derived as the dilation-group unitarity axis (Elie 5290 — exact). - The Weil positivity is the shared wall — UNMOVED by both. ★ My earlier claim that “BST derives the positivity Connes assumes” was a conflation of the *automatic* H^2 positivity with *Weil's* functional. Retracted. ☐ Honest bottom line: BST is *not* ahead of Connes on RH. BST brings a *derived operator* and a *derived critical line* (real, and more than Berry–Keating has); Connes brings the *arithmetic completion*. Neither has the Weil positivity / the object-is- ζ . The wall is the same for both.

The liability Connes does NOT have (and we must clear)

Connes' adelic object is built to be ζ (no risk of being the wrong function). BST's cone-zeta is a Koecher/Epstein zeta of a specific quadratic form, and Davenport–Heilbronn (1936) says such objects GENERICALLY have off-line zeros (class number > 1 / no Euler product). So BST carries a risk Connes doesn't: our own object may be a *counterexample* to its own RH. The Shimura-lift / class-number-1 (Eichler, favorable for quinary forms) is what clears it. Connes never has to.

Honest scorecard (referee-calibrated, $\leq 3/10 = \text{consensus}$)

- BST's derived Weil positivity: a genuine advance over Connes *if* it survives Cal's referee (is the sum-of-squares literally the Weil sum?). If yes, this alone is a real, publishable contribution — “the positivity Connes assumes is forced by measurement-as-commitment.”
- RH itself: neither Connes nor BST is at consensus. Both reduce it to a named gap; the gaps are *different* (Connes: positivity; BST: the object-is- ζ). BST's gap is arguably more tractable (a classical Shimura-lift with a favorable class-number) — but that is exactly what must be shown, not assumed.

The move this points at

The comparison says the leverage is entirely on (A) — is our cone-zeta ζ . So: the Shimura-lift is next (BST_Shimura_Lift_cone_form). If it lands, BST has both halves and the Connes comparison becomes “we derived his assumption and supplied our own arithmetic.” If it fails, we learn our object is an L-function cousin, and we say so.

— Keeper, 2026-08-16. Internal/gated. BST and Connes hold complementary halves (BST: derived positivity; Connes: the arithmetic). The Shimura-lift is the piece that would give BST both. DH is a liability Connes lacks; class-number-1 clears it. Nothing external.